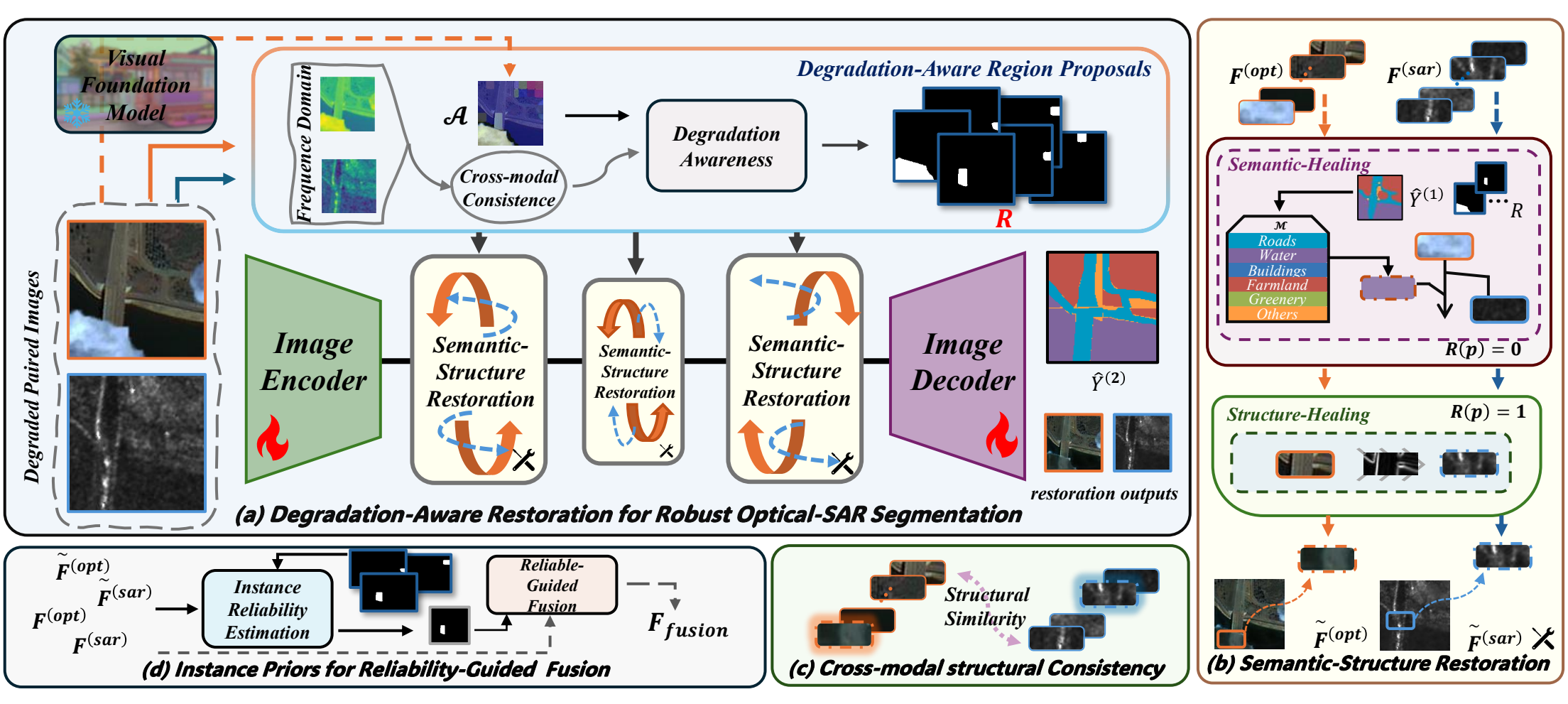

Visual Foundation Model
Degradation-Aware Region Proposals
Frequence Domain
$\mathcal{A}$
Cross-modal Consistence
Degradation Awareness
R
Degraded Paired Images
Image Encoder
Semantic-Structure Restoration
Semantic-Structure Restoration
Semantic-Structure Restoration
Image Decoder
$\hat{Y}^{(2)}$
restoration outputs
(a) Degradation-Aware Restoration for Robust Optical-SAR Segmentation
$F^{(opt)}$
$F^{(sar)}$
Semantic-Healing
$\hat{Y}^{(1)}$
R
$\mathcal{M}$
Roads
Water
Buildings
Farmland
Greenery
Others
$R(p) = 0$
Structure-Healing
$R(p) = 1$
$\tilde{F}^{(opt)}$
$\tilde{F}^{(sar)}$
(b) Semantic-Structure Restoration
$\tilde{F}^{(opt)}$
$\tilde{F}^{(sar)}$
$F^{(opt)}$
$F^{(sar)}$
Instance Reliability Estimation
Reliable-Guided Fusion
$F_{fusion}$
(d) Instance Priors for Reliability-Guided Fusion
Structural Similarity
(c) Cross-modal structural Consistency